

# Cambridge International AS & A Level

CANDIDATE  
NAME

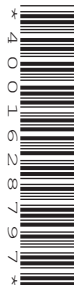
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## FURTHER MATHEMATICS

9231/21

Paper 2 Further Pure Mathematics 2

October/November 2022

**2 hours**

You must answer on the question paper.

You will need: List of formulae (MF19)

## INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.

## INFORMATION

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [ ].

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- 2 (a) Show that the system of equations

$$x - y + 2z = 4,$$

$$x - y - 3z = a,$$

$$x - y + 7z = 13,$$

where  $a$  is a constant, does not have a unique solution. [2]

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- (b) Given that  $a = -5$ , show that the system of equations in part (a) is consistent. Interpret this situation geometrically. [3]

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- (c) Given instead that  $a \neq -5$ , show that the system of equations in part (a) is inconsistent. Interpret this situation geometrically. [2]

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- 4 (a) Starting from the definitions of cosh and sinh in terms of exponentials, prove that

$$\cosh^2 x - \sinh^2 x = 1. \quad [3]$$

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- (b) Show that  $\frac{d}{dx}(\tan^{-1}(\sinh x)) = \operatorname{sech} x$ . [3]

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- (c) Sketch the graph of  $y = \operatorname{sech} x$ , stating the equation of the asymptote. [2]

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- (d) By considering a suitable set of  $n$  rectangles of unit width, use your sketch to show that

$$\sum_{r=1}^n \operatorname{sech} r < \tan^{-1}(\sinh n). \quad [3]$$

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- (e) Hence state an upper bound, in terms of  $\pi$ , for  $\sum_{r=1}^{\infty} \operatorname{sech} r$ . [1]

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$$\mathbf{A} = \begin{pmatrix} 2 & -3 & -7 \\ 0 & 5 & 7 \\ 0 & 0 & -2 \end{pmatrix}.$$

- (a) Find a matrix  $\mathbf{P}$  and a diagonal matrix  $\mathbf{D}$  such that  $\mathbf{A}^5 = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}$ . [7]

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- (b)** Use the characteristic equation of  $\mathbf{A}$  to show that

$$\mathbf{A}^4 = a\mathbf{A}^2 + b\mathbf{I},$$

where  $a$  and  $b$  are integers to be determined. [4]

This image shows a full page of a handwriting practice worksheet. It consists of approximately 20 horizontal rows. Each row is defined by two parallel dotted lines, creating a series of uniform gaps for letter height. The entire page is otherwise blank, with no margins, text, or other markings.

(b) Show that  $(1 + i \tan \theta)^k = \sec^k \theta (\cos k\theta + i \sin k\theta)$ , where  $\theta$  is not an integer multiple of  $\frac{1}{2}\pi$ . [2]

(c) By considering  $\sum_{k=0}^{n-1} (1 + i \tan \theta)^k$ , show that

provided  $\theta$  is not an integer multiple of  $\frac{1}{2}\pi$ . [5]

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- (d) Hence find  $\sum_{k=0}^{6m-1} 2^k \sin\left(\frac{1}{3}k\pi\right)$  in terms of  $m$ . [2]

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[illegible]

$$\theta \frac{dy}{d\theta} - y = \theta^2 \sin^{-1}(\theta - 1),$$

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This image shows a single page of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

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